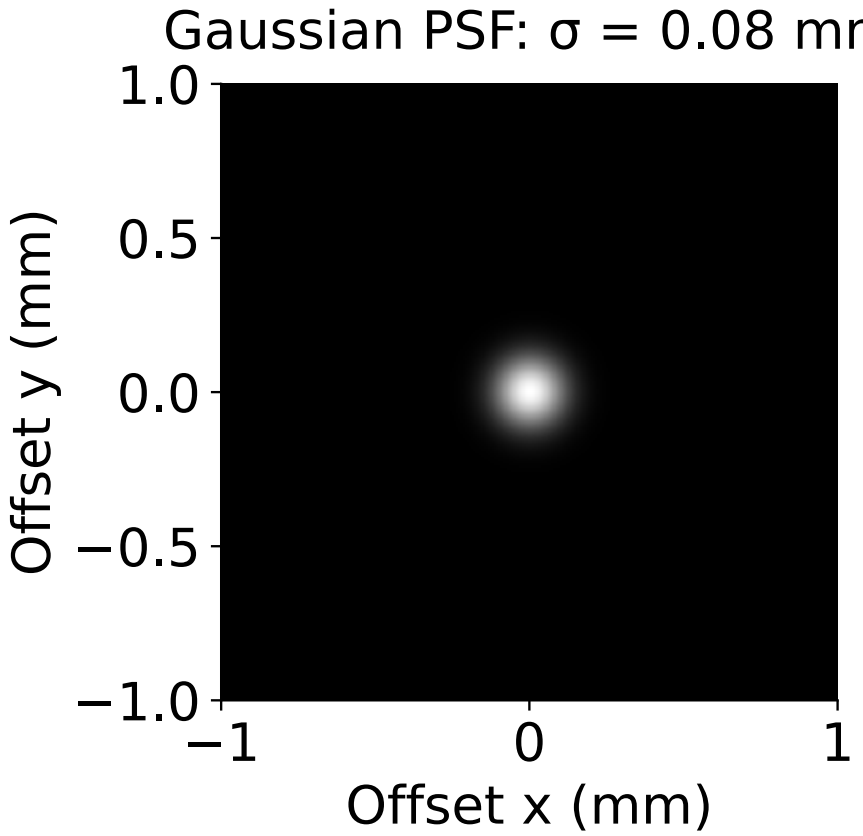
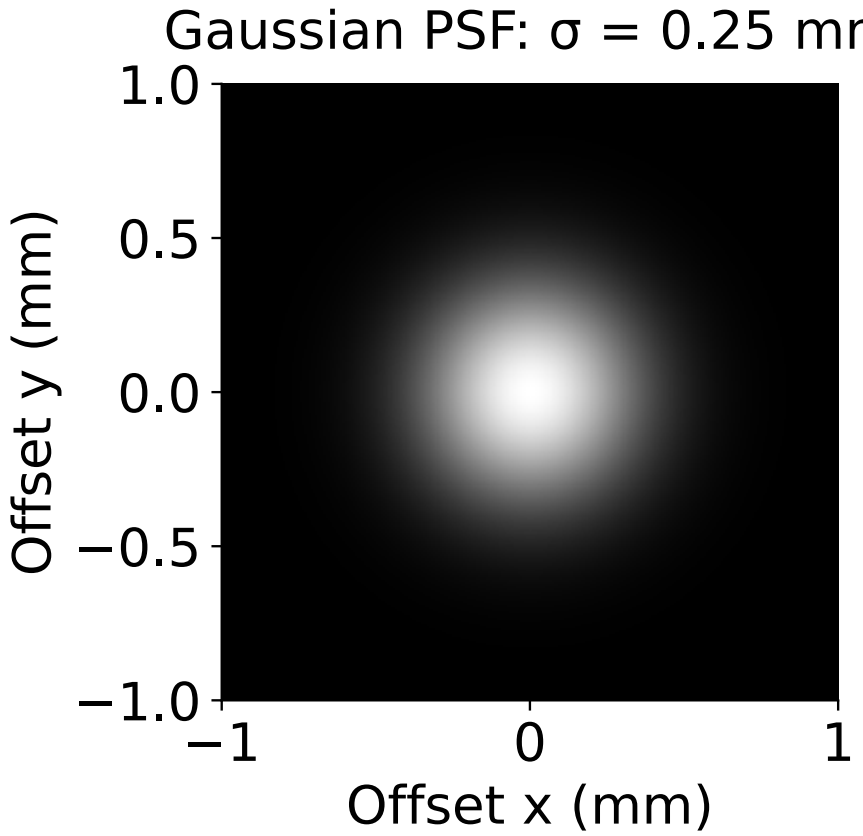
# System response

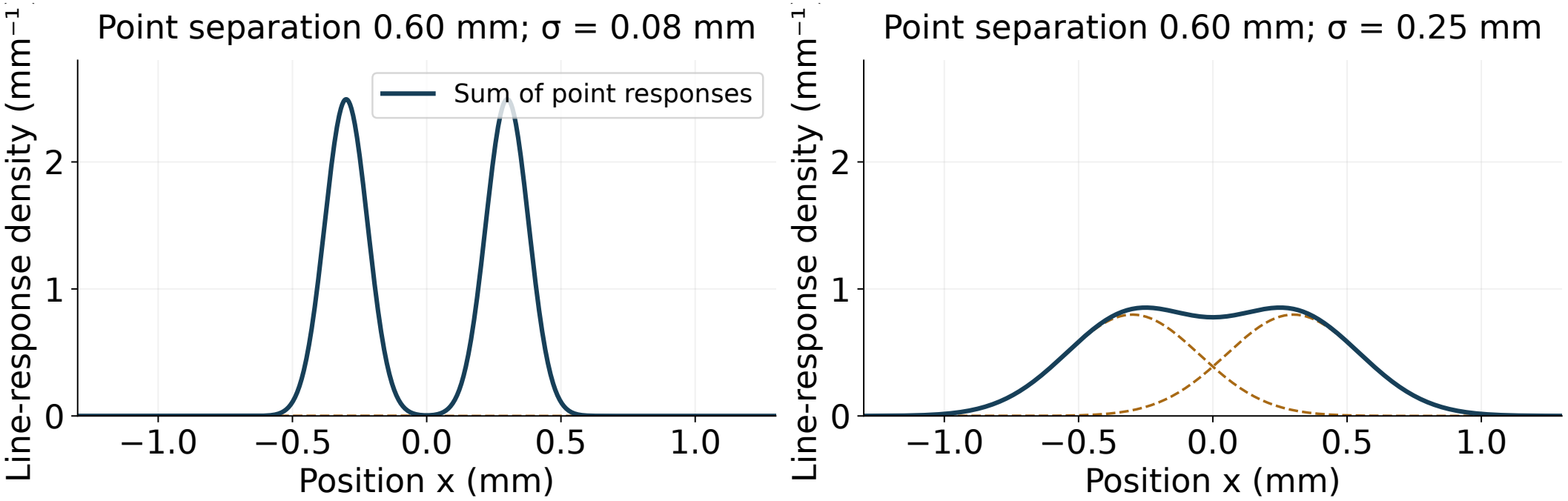
# A point becomes a spot




The **point-spread function (PSF)** describes the image of an ideal point input.

# Two nearby point inputs




A wider PSF makes the two objects overlap and hard to distinguish.

# Linear, shift-invariant imaging

For a **linear** system, weighted input signals produce the same weighted combination of output signals.

For a **shift-invariant** system, moving an input point shifts its response without changing its shape.

These assumptions allow one PSF to describe the system over the region of interest.

# Convolution and image formation

$$I_{\text{blur}}(x, y) = (I_0 * h)(x, y).$$

The system spreads each object point according to the PSF and adds the contributions.

$$I_{\text{blur}}(x, y) = \iint I_0(\xi, \eta)\, h(x - \xi, y - \eta)\, d\xi\, d\eta.$$

For an intensity-preserving blur model, $\iint h(x, y)\, dx\, dy = 1$.

# Object and image in the frequency domain

Use:

- $f(x, y)$ for the object,
- $h(x, y)$ for the impulse response,
- and $g(x, y)$ for the output image.

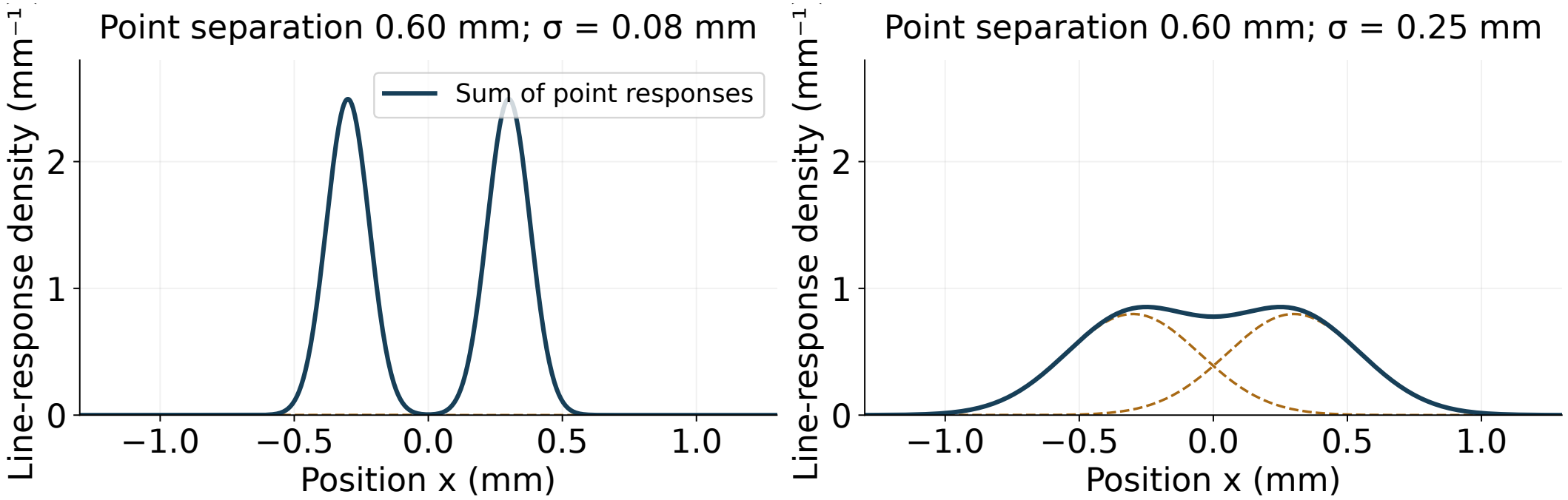
$$g = f * h \quad \Longleftrightarrow \quad G(u, v) = F(u, v)H(u, v).$$

The **transfer function** $H$ is the Fourier transform of $h$. Each object's frequency component is multiplied by the system's response at that frequency.

A constant input probes the zero-frequency response:

$$H(0, 0) = \iint h(x, y)\, dx\, dy.$$

# Two nearby point inputs




A wider PSF makes the two objects overlap and hard to distinguish.

# Gaussian PSF width

$$h(x,y) = \frac{1}{2\pi\sigma^2} \exp\left[-\frac{x^2+y^2}{2\sigma^2}\right].$$

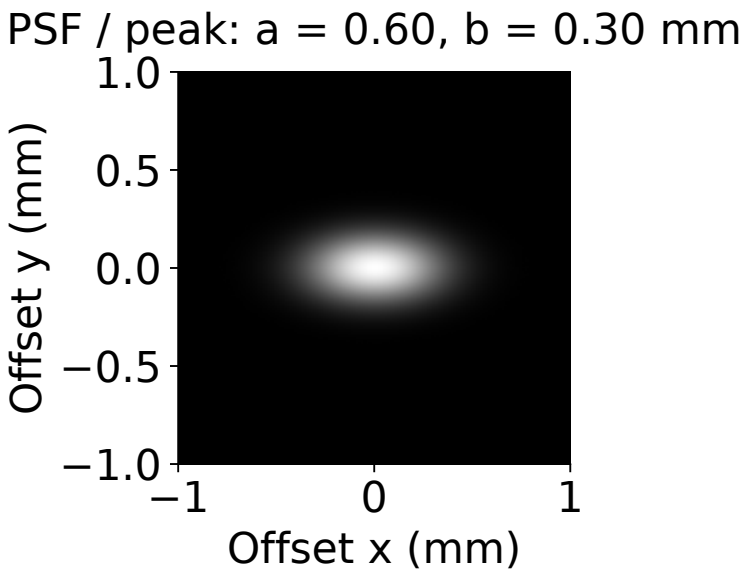
$\sigma$ sets the width of the spot and has units of mm.

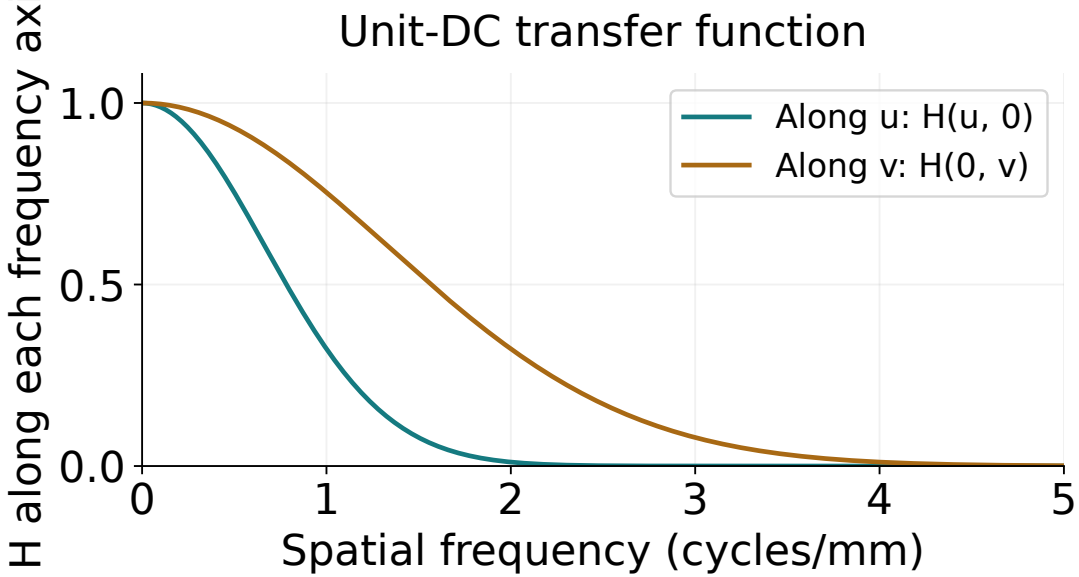
The **full width at half maximum (FWHM)** is the width across the central profile at half its peak value:

$$\text{FWHM} = 2\sqrt{2\ln 2}\,\sigma \approx 2.355\sigma.$$

**Example:** $\sigma = 0.10$ mm gives FWHM $\approx 0.236$ mm.

# Resolution can differ by direction




In this example, the PSF is wider along $x$. The response falls more rapidly with $u$, so the system preserves finer detail along $y$.

# Sinusoidal contrast

$$C = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}.$$

For $I(x) = \bar{I}[1 + C\cos(2\pi f x + \phi)]$, contrast is the amplitude divided by the mean.

The notebook target has $\bar{I} = 0.5$ and $C_0 = 0.9$, giving $I_{\max} = 0.95$ and $I_{\min} = 0.05$.

Blur can reduce contrast without changing the stripe frequency.

# Modulation transfer function

The **MTF** describes how much sinusoidal contrast the system preserves at each spatial frequency:

$$\mathrm{MTF}(f) = \frac{C_{\mathrm{out}}(f)}{C_{\mathrm{in}}(f)}.$$

For a normalized LSI blur model, it is the magnitude of the Fourier transform of the PSF.

An MTF of 0.5 at a particular frequency means the output sinusoid has half the input contrast at that frequency.

# MTF normalization and DC gain

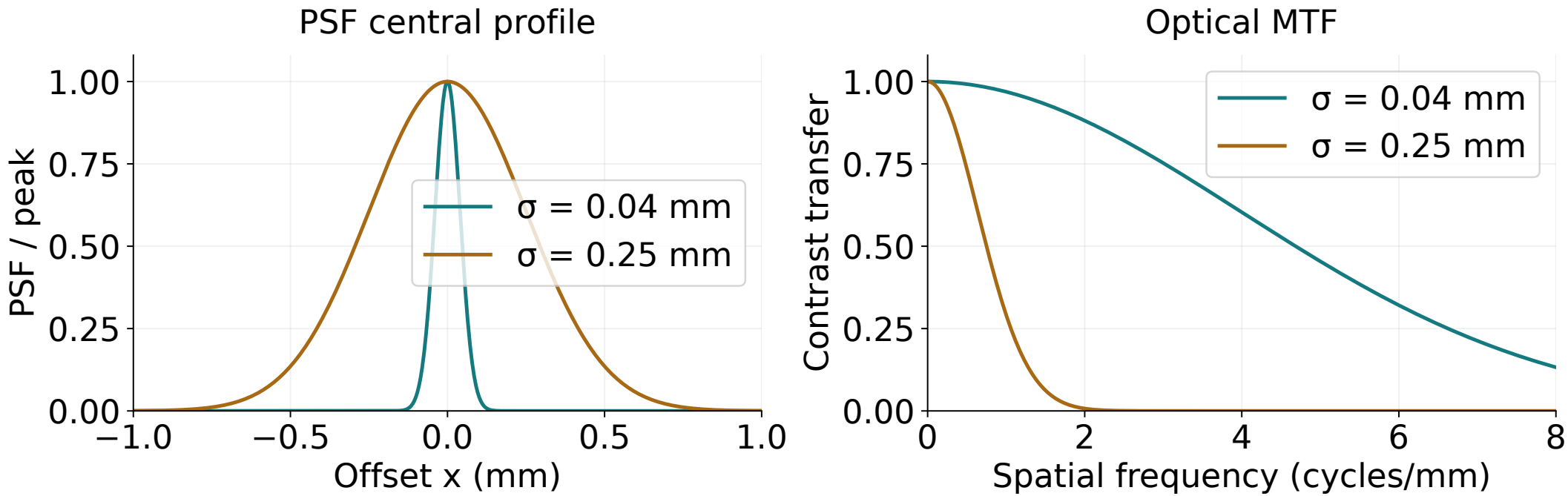
For an LSI system with positive DC gain,

$$\text{MTF}(u, v) = \frac{|H(u, v)|}{|H(0, 0)|}.$$

Dividing by the zero-frequency gain compares sinusoidal amplitude transfer with mean-intensity transfer. This gives contrast transfer and makes $\text{MTF}(0, 0) = 1$.

**Example:** if $H(0, 0) = 4$ and $|H(u_1, v_1)| = 2$, the MTF there is 0.5. An input contrast of 0.8 becomes 0.4.

# PSF width and contrast transfer




For a Gaussian, $H_{\text{opt}}(f) = \exp(-2\pi^2\sigma^2 f^2)$.

# Finite pixel size

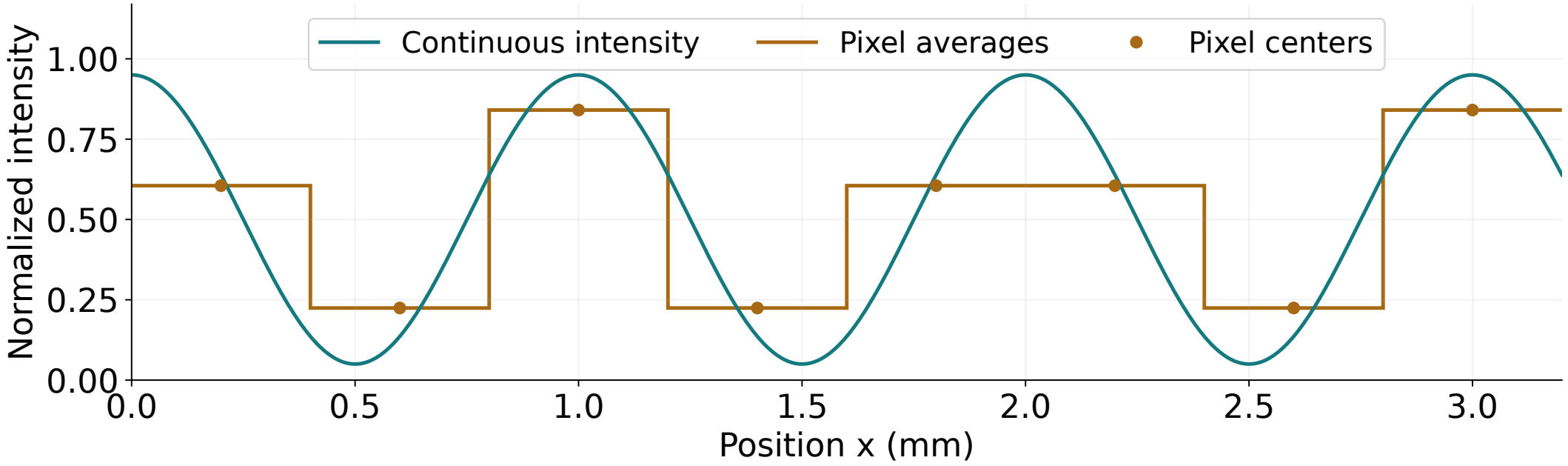
A pixel reports an average over its collecting area.

For vertical stripes and pixels of width and spacing $p$,

$$d_n = \frac{1}{p}\int_{x_n - p/2}^{x_n + p/2} I_{\text{blur}}(x)\, dx, \qquad x_n = (n + \tfrac{1}{2})p.$$

Larger pixels average over more of the spatial variation.

# Averaging and sampling




The step display assigns each recorded average to its pixel area.

# Combined response before sampling

For a sinusoid at frequency $f$, pixel averaging contributes

$$H_{\text{pixel}}(f) = \text{sinc}(pf), \qquad \text{sinc}(u) = \frac{\sin(\pi u)}{\pi u}.$$

The combined **presampling MTF** is

$$\text{MTF}_{\text{pre}}(f) = |H_{\text{opt}}(f) H_{\text{pixel}}(f)|.$$

Sampling adds a separate limit: $f_N = 1/(2p)$.

# Contrast loss below Nyquist

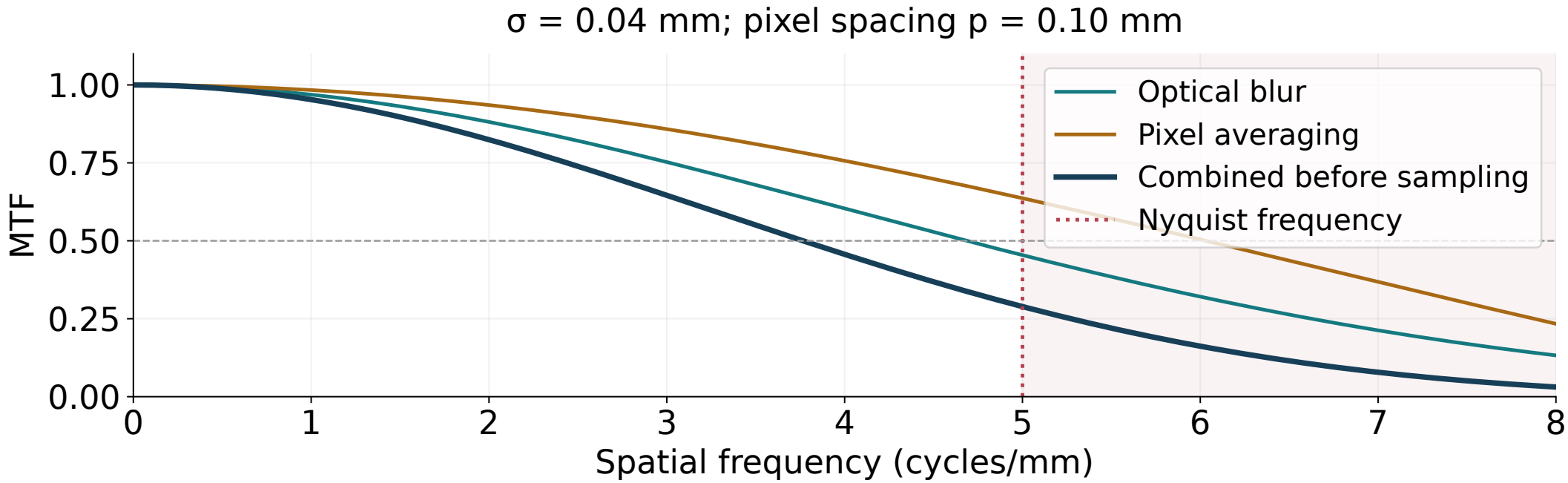
Take a sinusoid at $f = 1$ cycle/mm and pixels of width and spacing $p = 0.40$ mm.

$$f_N = \frac{1}{2p} = 1.25 \text{ cycles/mm}, \qquad |H_{\text{pixel}}(f)| = |\operatorname{sinc}(0.40)| \approx 0.757.$$

The frequency is below Nyquist, yet pixel averaging preserves only about 76% of the continuous sinusoidal contrast.

- **Averaging:** nearby bright and dark intensities contribute to one pixel value.
- **Aliasing:** different continuous frequencies produce the same samples.

# An MTF curve and a sampling limit




**MTF50** is the first frequency where the combined MTF falls to 0.5.

# Noise and Signal-to-Noise Ratio (SNR)

When imaging photons or particles the incoming signals follows a Poisson distribution. For $N$ average signal photons:

- The standard deviation (noise) is $\sigma = \sqrt{N}$
- The mean (average) is $\mu = N$
- The SNR is signal/noise = $\frac{\mu}{\sigma}$.
  - For Poisson noise $SNR = \frac{N}{\sqrt{N}} = \sqrt{N}$.

This is why more light makes for a better image!

# Reading and resources

Prince and Links, *Medical Imaging: Signals and Systems*: LSI imaging, point-spread functions, and image quality.

- Michigan EECS 516: Medical Imaging Systems
- MIT RES.6-007: convolution and Fourier-transform review
- Course notebook: **01_sampling_resolution.ipynb**, investigation 1.

# Backup

**BACKUP**

# Separable Gaussian responses

If $h(x, y) = h_x(x)h_y(y)$, its transform factors:

$$H(u, v) = H_x(u)H_y(v).$$

For $a, b > 0$,

$$h(x, y) = Ae^{-\pi[(x/a)^2+(y/b)^2]} \quad \Longrightarrow \quad H(u, v) = Aab\, e^{-\pi[(au)^2+(bv)^2]}.$$

**Method:** transform each one-dimensional Gaussian, include its scale factor, and multiply. Check $H(0, 0)$ against the integral of $h$.